NICHOLAS SOBCHAK

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Portfolio | LinkedIn | GitHub | U.S. Citizen

EDUCATION

University of Maryland, Baltimore County (UMBC) Expected May 2027
Bachelor of Science in Computer Science
Relevant coursework: Neurosymbolic AI, Database Management Systems, Data Visualization

Carroll Community College May 2025
Associate of Arts in Computer Science

TECHNICAL SKILLS

Languages: Python, C++17, Java, SQL, TypeScript, JavaScript
Systems & Tools: Linux, Git, Docker, CMake, GitHub Actions, Valgrind
Testing: pytest, Catch2, JUnit
Backend & Data: Spring Boot, JPA, PostgreSQL, SQLite, DynamoDB
Cloud: AWS Lambda, API Gateway, Cognito, SAM, CloudFormation, boto3
Frontend: Angular, React, HTML, CSS

EXPERIENCE

Founding Software Developer – ZipTip Dec 2025 – Present
Remote / Westminster, MD

- Designed a serverless scan API with AWS Lambda, API Gateway, and DynamoDB, implementing authentication, custom-domain routing, and ID-to-account redirects for a real-time tipping platform.
- Built AWS SAM CI/CD pipelines with OIDC authentication, automated Route 53 DNS updates, and rollback support for development and production environments.
- Built unit and integration tests with pytest for backend services and automated AWS SAM deployments.
- Developed account authentication and authorization services using Amazon Cognito; integrated backend APIs with Python and boto3.

TECHNICAL PROJECTS

QuickQuill – *C++17, Java, Spring Boot, SQLite, PostgreSQL, Docker*

- Built a trie-based C++ dictionary engine supporting lookup, spell-check suggestions, and autocomplete across a 1.28M-word dictionary.
- Used mutex synchronization for an LRU cache and thread-local SQLite connections for concurrent lookups; managed trie nodes and database handles with C++ smart pointers.
- Integrated the native engine with a Spring Boot REST API through Java's Foreign Function & Memory API and an `extern "C"` interface.
- Secured account endpoints with validated JSON credential bodies, BCrypt password hashing, and PostgreSQL-backed sessions; added request-validation and credential-leak regression tests.

On The Money – *C++17, Java, Spring Boot, PostgreSQL, React, Docker*

- Integrated a C++ Monte Carlo engine with a Spring Boot REST API to run 10,000 simulations per request for retirement and portfolio growth projections.
- Protected Plaid access tokens at rest with AES-GCM encryption using Spring Security; kept token exchange and storage in the backend.
- Integrated Plaid Link for bank-account connections and transaction synchronization; deployed a Dockerized Java API and React PWA backed by PostgreSQL.